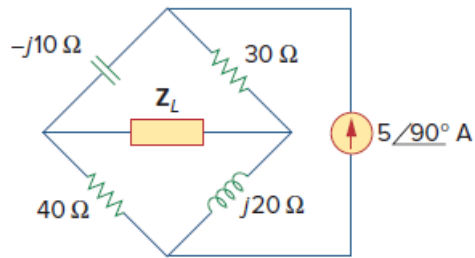


Problem

Calculate the value of $\mathbf{Z_L}$ in the circuit of Fig. 11.48 in order for $\mathbf{Z_L}$ to receive maximum average power. What is the maximum average power received by $\mathbf{Z_L}$?

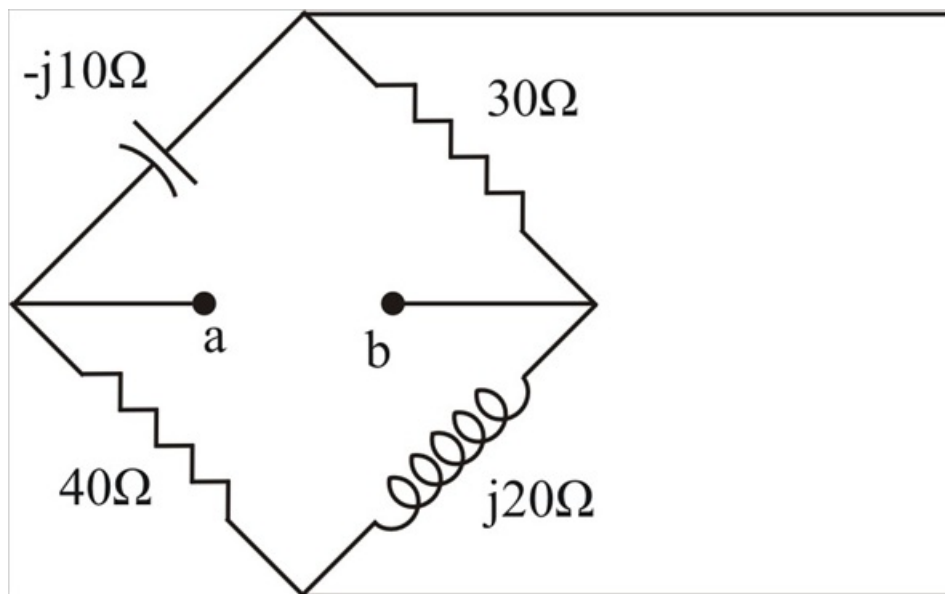
Figure 11.48:



Step-by-step solution

Step 1 of 5

We find RTH at terminals a – b following fig(a).



Step 2 of 5

$$Z_{TH} = -j10 + 30 \parallel j20 + 40$$

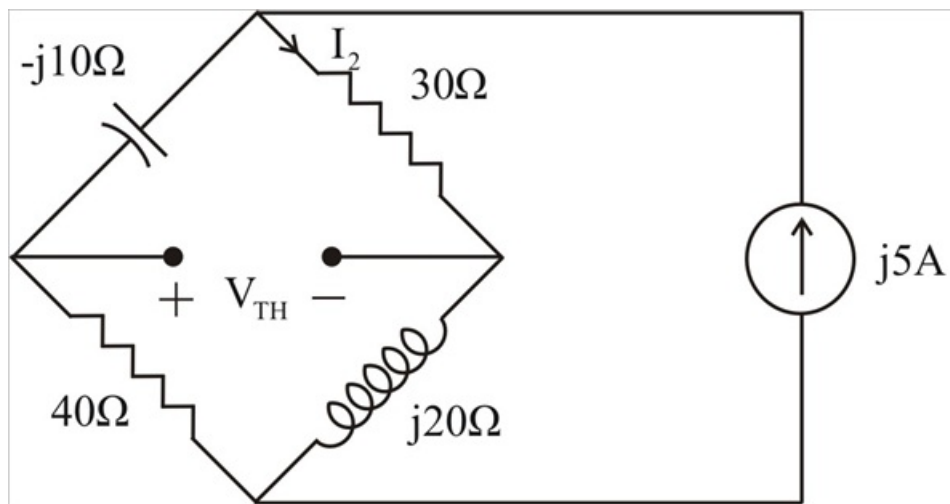
$$Z_{TH} = \frac{(30 - j10)(40 + j20)}{70 + j10}$$

$$Z_{TH} = 20 \Omega$$

$$Z_L = 20 \Omega$$

Step 3 of 5

We obtain V_{TH} from fig(b).



Step 4 of 5

Using current division,

$$I_1 = \left(\frac{30 + j20}{70 + j10} \right) (j5)$$

$$I_1 = -1.1 + j2.3 \text{ A}$$

$$I_2 = \left(\frac{40 - j10}{70 + j10} \right) (j5)$$

$$I_2 = 1.1 + j2.7 \text{ A}$$

Step 5 of 5

$$V_{TH} = 30I_2 + j10I_1$$

$$V_{TH} = 10 + j70$$

$$V_{TH} = 70.711 \angle 81.87^\circ \text{ V}$$

$$P_{\max} = \frac{|V_{TH}|^2}{8Z_L}$$

$$P_{\max} = \frac{5000}{(8)(20)}$$

$$\therefore \boxed{P_{\max} = 31.25 \text{ W}}$$